

HVAC parts list - 312 Sudden Valley Dr, Bellingham WA 98229 (Sudden Valley, unincorporated Whatcom County)

SFR2025-00185 model 312-suddenvalley-hvac printed 2026-09-25 ducted heat 18,306 Btuh cooling 0.97 t 1 outdoor 0 heads 1 ducted units 13 registers line sets 26 ft duct 337 ft vent 285 ft 14 days on site
Design 19 F / 78 F (sudden valley: WSEC Table RC-1 Bellingham IAP 19 F / 78 F [S]; MJ8 Table 1A 23 F / 76 F [V] (Sudden Valley is ~8 mi from the airport, 300-700 ft up - STAND-IN on the Bellingham row)). Foundation crawl, line sets concealed, ducts crawl_vented, ventilation erv.
Quantities only - the prices are set on the price list in the app.

Loads

heating 18,306 Btuh (12.2 Btuh/sf) · cooling 10,763 + 902 latent · WSU sheet 24,190 Btuh · infiltration 1.5 ACH50 -> 0.135 ACH at design · ventilation 45.0 cfm (erv)

room	storey	sf	heat Btuh	cool Btuh	glass sf	ext wall ft
LIVING ROOM	upper	254	4,630	3,501	158	59
BDRM 2	main	127	1,491	906	45	25
MASTER BATH	upper	58	1,403	588	50	22
FOYER	main	89	1,275	431	28	27
BATH	main	103	1,274	404	37	22
DINING	upper	121	1,250	733	50	13
MSTR BEDROOM	upper	140	1,213	864	28	21
KITCHEN	upper	138	1,141	2,070	28	18
HALL	main	23	821	509	18	18
LANDING	upper	41	617	112	16	10
HALL	main	16	604	132	19	10
HALL	main	22	576	72	8	14
BUILT IN CLOSET	upper	34	543	157	25	5
BDRM 3	main	125	356	235	0	6
CLST	main	16	167	2	0	5
LAUNDRY	main	12	142	2	0	4
CLST	main	12	16	0	0	0
CLST	main	11	15	0	0	0
CLST	main	10	14	0	0	0

The system

unit house - ducted: heat 18,306 Btuh, capacity at design 18,803 Btuh
outdoor eq_hp_24 24k (central) on the S wall, at the electrician's Heat pump condenser (LIVING ROOM, electrical/jobs/suddenvalley/devices.json); 26 ft of line set
indoor eq_ahu_24 24k in MECHANICAL ROOM UNDER STAIRS (main), 588 cfm

register room	storey	cfm	duct
BDRM 2	main	43	5 in
FOYER	main	37	5 in
BATH	main	37	5 in
HALL	main	58	6 in
BDRM 3	main	30	5 in
MASTER BATH	upper	41	5 in
DINING	upper	36	5 in
KITCHEN	upper	49	5 in
KITCHEN	upper	49	5 in
LIVING ROOM	upper	83	6 in
LIVING ROOM	upper	83	6 in
MSTR BEDROOM	upper	41	5 in
LANDING	upper	30	5 in
return	storey	cfm	duct
FOYER	main	205	8 in
LANDING	upper	413	12 in

grille
14x20
14x20

Runs (measured, before waste)

system / size	LF
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dryer 4	4
erv_exhaust 4	33
erv_exhaust 5	46
erv_outdoor 6	26
erv_supply 4	93
erv_supply 6	65
exhaust 4	5
exhaust 6	8
hood 6	6
refrigerant 0.375x0.625	26
return 12	17
return 8	28
supply 12	9
supply 5	158
supply 6	42
trunk 12	50
trunk 8	33

Equipment

Equipment

item	measured	order	unit	note
Air handler - variable speed, 24k (2 ton)	1	1	EA	air handler 24k in MECHANICAL ROOM UNDER STAIRS (main; job.json (main/MECH))
Backup heat strip 5 kW (controlled per WSEC R403.1.2)	1	1	EA	backup / defrost tempering 5 kW (heat-pump shortfall at design 0 Btuh; Manual S: <= 5 kW while it is under 15,
ERV / HRV - balanced whole-house, 50-150 cfm (SRE per the credit)	1	1	EA	unit house: ERV / HRV - 45 cfm continuous (balanced, ducted to every habitable room: Csystem 1.0), SRE >= 0.7
Heat pump outdoor unit - centrally ducted 24k (2 ton), cold-climate variable capacity	1	1	EA	ducted heat pump outdoor unit 24k (~18,803 Btuh at 19 F), at the electrician's Heat pump condenser (LIVING ROO
Thermostat - programmable, heat pump with aux-heat lockout (WSEC R403.1.1 / R403.1.2)	1	1	EA	WSEC R403.1.1 programmable; heat-pump aux lockout R403.1.2

Rough-in

Condensate

item	measured	order	unit	note
Condensate line 3/4 in PVC (or 5/8 vinyl with the line set)	9.5	11	LF	air handler condensate to an approved discharge
Condensate trap + cleanout tee	1	1	EA	trap + cleanout tee

Ductwork

item	measured	order	unit	note
Duct hanger strap / saddle (flex every 4 ft, rigid every 10 ft)	68	68	EA	trunk
Duct insulation R-8 wrap (rigid duct outside the envelope, WSEC R403.3.1)	33.2	37	LF	main trunk in the vented crawl - R-8 (WSEC R403.3.1)
Duct mastic (gal) + UL-181 foil tape	4	4	EA	mastic by the gallon per 100 ft of duct
Flex duct R-8 5 in	157.6	166	LF	BDRM 2: 43 cfm · FOYER: 37 cfm · BATH: 37 cfm · BDRM 3: 30 cfm · MASTER BATH: 41 cfm · DINING: 36 cfm
Flex duct R-8 6 in	42.2	45	LF	HALL: 58 cfm · LIVING ROOM: 83 cfm
Register boot (4x10 / 4x12 floor or 6x10 ceiling)	13	13	EA	
Return duct 12-16 in (panned joist not allowed for a new return)	45.4	48	LF	main central return 8 in, 205 cfm, 14x20 filter grille · upper central return 12 in, 413 cfm, 14x20 filter gri
Supply + return plenums (fabricated, sealed, lined where required)	1	1	EA	supply + return plenums at the unit
Supply trunk - galvanized rectangular / 10-14 in round (extended plenum)	94	99	LF	main: 8 in equivalent, 205 cfm - reducing past 24 ft · upper: 12 in equivalent, 413 cfm - reducing past 24 ft
Takeoff / starting collar with balancing damper	13	13	EA	
Transfer grille pair / jumper duct (closed-door pressure relief)	3	3	EA	BDRM 2: 43 cfm needs ~24 sq in free area at 3 Pa - transfer grille pair or jumper duct · BDRM 3: 30 cfm needs

Refrigerant piping

item	measured	order	unit	note
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Anti-vibration pads (set of 4)	1	1	EA	
Communication cable 14/4 stranded (outdoor unit to each indoor unit)	26.5	30	LF	with the line set
Equipment pad - composite, sized to the outdoor unit	1	1	EA	outdoor unit pad
Line set 3/8 x 5/8 insulated	26.5	30	LF	24k indoor unit, 26 ft
Outdoor unit stand / wall bracket (snow and drainage clearance)	1	1	EA	snow / drainage stand
Wall penetration - 3 in sleeve, sealed and flashed	1	1	EA	rim penetration for the line set

Support + protection

item	measured	order	unit	note
Fire caulk / sealant (tube)	1	1	EA	fire caulk / sealant at the penetrations

Ventilation and exhaust

Ventilation

item	measured	order	unit	note
Bath fan exhaust duct 4 in insulated flex (50 cfm fan: <= 25 ft, Table M1505.4.4.2)	4.7	6	LF	MASTER BATH: 50 cfm, 5 ft, 2 elbows -> 4 in flex (Table M1505.4.4.2 allows 25 ft)
Dryer duct 4 in 90 elbow	2	2	EA	
Dryer duct 4 in rigid metal, smooth wall (M1502.4.2)	3.6	4	LF	from the electrician's dryer outlet in LAUNDRY: 4 ft
Dryer vent box (recessed)	1	1	EA	
Dryer wall cap with damper, no screen (M1502.3)	1	1	EA	
ERV duct 5-6 in (supply to bedrooms / living, exhaust from baths / kitchen / laundry)	237.1	261	LF	supply trunk, main · supply BDRM 2 · supply BDRM 3 · supply trunk, upper - in the floor cavity (vaulted storey
ERV intake / exhaust hood 6 in (intake >= 10 ft from any exhaust, R303.5 / M1505.2)	2	2	EA	outdoor-air intake - 12 ft apart on the N wall (M1504.3: exhaust >= 10 ft from an intake) · exhaust outlet - 1
ERV outdoor-air / exhaust duct, insulated with vapour jacket (to the hoods)	25.8	29	LF	ERV outdoor-air intake to its wall hood (insulated, vapour jacket) · ERV exhaust outlet to its wall hood (insu
ERV supply / exhaust grille or valve 4-6 in	7	7	EA	BDRM 2 · BDRM 3 · DINING · LIVING ROOM · MSTR BEDROOM · BATH
Exhaust duct 6 in insulated (80-110 cfm fans, whole-house fan: Table M1505.4.4.2)	7.8	9	LF	BATH: 110 cfm, 8 ft, 3 elbows -> 6 in flex (Table M1505.4.4.2 allows 15 ft)
Range hood duct 6-10 in rigid, sealed (M1503.3)	6.4	7	LF	6 in rigid for a 300 cfm hood at the electrician's range outlet (electrical/jobs/suddenvalley/devices.json) (M
Range hood wall / roof cap with damper	1	1	EA	6 in wall cap with damper
Wall / roof cap 6 in with backdraft damper	1	1	EA	BATH: wall cap with backdraft damper
Wall cap 4 in with backdraft damper	1	1	EA	MASTER BATH: wall cap with backdraft damper

Trim

Registers + grilles

item	measured	order	unit	note
Filters MERV 8-13 (first set)	1	1	EA	first filter set
Floor register 4x10 / 4x12	13	13	EA	BDRM 2 · FOYER · BATH · HALL · BDRM 3 · MASTER BATH
Return / filter grille 20x25 (or 14x30)	2	2	EA	FOYER (main): 14x20 at <= 300 fpm · LANDING (upper): 14x20 at <= 300 fpm

Start-up, tests and design

Commissioning

item	measured	order	unit	note
Air balance of supply registers (per ducted system)	1	1	EA	register balance
Duct leakage test, total cfm25 (WSEC R403.3.5) - certified tester	1	1	EA	ducts outside the thermal envelope (vented crawl): total leakage test (WSEC R403.3.5)
Equipment labels, WSEC certificate data, owner hand-over	1	1	EA	per dwelling unit
Load calculation + equipment selection for the permit (Manual J / S or the WSU sizing form)	1	1	EA	room-by-room loads for the permit (this takeoff is the draft)
Start-up, charge verification, controls set-up (per outdoor unit)	1	1	EA	24k system
Ventilation flow test - whole-house + local exhaust (WA M1505.4.3 / M1505.4.4)	1	1	EA	unit house: whole-house 45 cfm + every local exhaust measured at the grille (WA M1505.4.3 / M1505.4.4)

Refrigerant piping

item	measured	order	unit	note
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Nitrogen pressure test + evacuation to 500 micron (per system)	1	1	EA	per refrigerant system
Refrigerant - additional charge beyond the pre-charge (lb)	0.1	0.1	lb	1 oz past the pre-charge [U: the maker's oz / ft]

The checker

DATA plan set: credit 2.2 (heat-recovery ventilation) AND a whole-house exhaust fan are both on the set -> hrv (if the ERV/HRV is the whole-house system the fan is local exhaust only; if the fan is the whole-house system the credit fails - ask the designer)

DATA design temperature: 19 F is a stand-in (WSEC Table RC-1 Bellingham IAP 19 F / 78 F [S]; MJ8 Table 1A 23 F / 76 F [V] (Sudden Valley is ~8 mi from the airport, 300-700 ft up - STAND-IN on the Bellingham row)) - pull WSEC Table RC-1 for sudden valley before this goes out

note: tracer fragments merged into one space (same name, touching): CLST (main: 4 pieces, 49 sf); HALL (main: 3 pieces, 61 sf)

note: bath fans: 2 read from the electrical skill (electrical/jobs/suddenvalley/devices.json) - the electrician supplies and wires them; HVAC ducts and caps each one

note: unit house: ERV supplies 5 habitable rooms and exhausts 2 wet rooms on a trunk-and-branch layout - 237 ft of 4-6 in duct

note: equipment must meet HSPF2 >= 9.4 (the WSEC credit); design temperature 19 F <= 23 F: a NEEP-listed cold-climate variable-capacity unit (the 3.6 note)

note: ducts in the vented crawl are outside the thermal envelope: R-8 on every duct, mastic-sealed, and the total duct leakage test before final

note: refrigerant: a split system installed from 1/1/2026 is an A2L (R-454B / R-32) unit under the EPA Technology Transitions rule; R-410A equipment is out. A2L units carry leak detection / mitigation per their listing - it rides with the equipment line

Labour and inspections

crew of 2: set 15.4 h / 3 d, rough 51.4 h / 9 d, trim 2.3 h / 1 d, startup 6.5 h / 1 d - 14 days at 6.25 h. crew minutes are PLACEHOLDERS (no task-level install hours were found: references/research-hvac-market.md 3.3) - set them from a sub or the crew

Mechanical rough-in (cover) - after framing, before insulation; line sets pressure-tested and in place; ducts, boots, exhaust ducts and caps; firestops at the garage separation

Duct leakage test - rough-in or final; total leakage at 25 Pa per WSEC R403.3.5 - waived only when every duct and the air handler are inside the envelope

Ventilation flow test - before final; whole-house rate and every local exhaust measured (WA M1505.4)

Final mechanical - equipment set, started, labelled; equipment listing / HSPF2 on the certificate, controls, condensate, clearances

permit: Whatcom County PDS residential mechanical fee schedule - NOT FOUND in the 9/25 research (references/research-hvac-code.md 8): price it from the county sheet